

DEM investigation of the interaction between rapid dry granular flow and dual-slit structures

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ABSTRACT

Granular flow are considered as the one of the most hazardous geological disasters in mountainous area. Multi-slit structure including one or more slits could partially block the granular flow and thus allow downstream flow at a reduced rate, have played a vital role in granular flow geohazard mitigation. This study conducted a series of discrete element method (DEM) models to investigate the interaction between dry granular flow and multi-slit structures (dual-slit structures in this study), and characterize the regulation functions of multi-slit structures. The dynamics process of flow-slit structure interaction and the effects of granular arches in multi-slit structures has been explicitly analyzed. Based on the analyses, a further modified Beverloo law considering the contraction behaviors is proposed to quantify the relationship between the mass outflow rate and relative slit size. The simulation findings show that both the slit size, slit numbers and slit separation have a significant impact on the regulating functions, such as pile-up height, mass outflow rate, and grain-trapping efficiency. In addition, it was found that, the shearing effect of the particles between the two nearby slits, the forming of the granular arches and the Froude characteristics might together influence the flow-slit interaction process, a key value of 4 for relative slit separation under a lower Froude number (5.86, in this study) was found to offer a reference for multi-slit structure scientific optimization design.

1. Introduction

Granular flow are one of the most common and hazardous geological disasters in mountainous areas due to their enormous volume (up to 10^6 m³), rapid velocity (usually >10 m/s), long runout distance, and poor temporal prediction (Albaba et al., 2018; Brighenti et al., 2013; Huang and Zhang, 2020; Hung et al., 2014; Iverson, 1997; Iverson et al., 2016, 2011; Pudasaini and Miller, 2013; Pudasaini and Krautblatter, 2021). Therefore, granular flows pose a catastrophic threat to downstream communities and infrastructure (Cliff and Dilip, 2014; Iverson, 2015; Iverson et al., 2016; Leonardi, 2020). On 7 August 2010, for example, a disastrous rainstorm-triggered debris flow disaster in Zhouqu County, Gansu Province, China killed 1765 people and damaged over 5500 dwellings (Tang et al., 2011). Consequently, scientific and engineering communities are increasingly concerned with the design of effective measures to prevent granular flow disasters.

To mitigate such destructive hazards, different mitigation measures, such as closed check dams (Chen et al., 2019; Faug et al., 2012; Faug,

2015), braking mounds (Kattel et al., 2018), flexible Barrier (Albaba et al., 2017; Song et al., 2019), and slit structures are often installed in the flow channel to fully trap granular flows. However, when the entire channel width is blocked, the channel's capacity of holding granular flows is limited by the channel's scale. When the internal capacity is filled, the closed check dam will lose its interception function until it is cleared out, which is time-consuming and laborious (Goodwin, 2019; Goodwin and Choi, 2020). Therefore, the slit structure, especially multi-slit structure have been increasingly used as a prevention and mitigation measurement against granular flow, because of not only (i) preventing the structure from becoming fully filled up, but also (ii) reducing the flow rate downstream of the granular flow (Chen et al., 2021; Choi et al., 2016; Leonardi et al., 2019). However, the apertures of slit structures may induce the clogging of grains in granular flow fronts, which form the dynamic granular arches and further complicate the interaction mechanism (Lai et al., 2020; Zhou et al., 2020). Many small-scale experimental studies and numerical modeling have been carried out in order to investigate the granular flow impact issues on slit structures.

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Choi et al. (2016), for example, conducted flume tests to evaluate the interaction between dry coarse-grained flows and slit structures. Their results revealed that, the grain-trapping efficiency depends on the relative slit size. Zhou et al. (2019) investigated the effect of the Froude characteristics and relative slit size on the run-up heights of granular flows using the discrete element method (DEM). Furthermore, Leonardi et al. (2019) studied the granular flow impact load as a function of particle size, slit size, and channel inclination using a physical flume test and DEM modeling. The results reveal that the impact pressure around the slit is almost twice as high as the impact pressure elsewhere. Marchelli et al. (2020) further investigated the clogging process during flow-slit structures. Huang et al. (2021) proposed a multiple-indicator-based framework to assess the effect of the grain size and baffle shape on baffle arrays.

General speaking, prior investigations, while effective in gaining a fundamental knowledge of flow-slit-structure interactions, were restricted to single-slit structures. However, the hopper-discharge researches have justified that multi-slit cases show some distinct characteristics in flow rate (Kunte et al., 2014; Zhang et al., 2016). Multi-slit structures, according to Goodwin and Choi (2020), are a more resilient alternative for engineering design since they can boost grain trapping efficiency while lowering peak outflow rate. Thus, it can be inferred that geometric features (i.e. slit width (s), slit separation (L) and slit numbers (N_s)) of slit structure should also be considered for designing slit structures. Therefore, the functions of multi-slit structures for intercepting and regulating rapid dry granular flows was systematically investigated in this study by comparing with single-slit structures. To the best of our knowledge, despite continuum-based methods, such as depth-averaged continuum models (George and Iverson, 2014; Hung et al., 1984; Iverson and George, 2014), multi-phase mass flow model (Mergili et al., 2020; Pudasaini and Mergili, 2019), smoothed particle hydrodynamics and material point method (Choi et al., 2021; Wang et al., 2018) have been developed and widely used to investigate the behavior of landslide and debris flow, they are generally considered when large volumes are involved (Kattel et al., 2018). And this method has to be given explicitly by macroscopic constitutive laws (Gracia et al., 2019). On the other hand, the DEM requires an accurate definition of particle shapes, together with quite long computational durations. As a consequence, DEM turns out to be more appropriate for “small or medium sized volumes” or more generally when the flow bodies are prone to show discrete characteristics. Moreover, the complex contact mechanics at multiscale in granular flows are naturally well captured by the DEM (Gong et al., 2021; Hu et al., 2020). Based on the analysis above, for the dry granular flow of interest and small sized volumes involved in this manuscript, the DEM approach was utilized to explore the interaction of granular flow with multi-slit structures (taking dual-slit structure as an example) in this paper (Cundall and Strack, 1979).

The remainder of this paper is structured as follows. First, the laboratory flume tests performed by Jiang and Towhata (2013) was considered as a reference to design a DEM model in this study. After that, the model was calibrated by comparing the granular flow progress and the normal impact load's time history curves. The dynamics process of flow-slit structure interaction and the effects of granular arches during flow-slit structure interaction are analyzed through combining numerical results and previous observations reported in the literature. The regulation functions of multi-slit structures relationship on pile-up height, mass flow rate, grain-trapping efficiency, and energy breaking capacity are detailed described. Finally, based on the above-mentioned results, several discussions and suggestions are provided for improving the rational design of multi-slit structures to prevent granular flow.

2. Numerical simulation

2.1. DEM theory

EDEM (DEM solutions, 2018) is a popular tool for dealing with

granular flows problems because of its capable of capturing particle movements and force chain networks at the microscale (Huang et al., 2021; Jiang et al., 2018). In this research, granular flows were modeled as an ensemble of rigid monodisperse spheres, and the structure was modeled using wall elements without deformation. The following equations govern the motions of particle i :

$$m_i \left(\frac{d\vec{v}_i}{dt} - g \right) = \sum_{j=1}^{n-1} (\vec{F}_{ij}^n + \vec{F}_{ij}^t) \quad (1)$$

$$I_i \frac{d\vec{\omega}_i}{dt} = \sum_{j=1}^{n-1} \vec{M}_{ij} \quad (2)$$

where m_i is the particle mass, g is the gravity acceleration; $\vec{v}_i$ is the translational velocities, $\vec{\omega}_i$ is angular velocities; I_i is the moment of inertia; $\vec{F}_{ij}^n$ and $\vec{F}_{ij}^t$ are the normal and tangential contact forces between the particles or between the particle and the structure; $\vec{M}_{ij}$ is the torque.

In this paper, Hertz–Mindlin (no slip) model are selected to calculate the particle contact force (Tsuji et al., 1992). This model has been widely implemented in the studies of dry granular flow (Zhang and Huang, 2022; Zhou et al., 2020). The normal contact force is expressed as follows.

$$F_n = F_n^c + F_n^d = \frac{4}{3} E^* \sqrt{R^*} \delta_n^{3/2} - 2 \sqrt{\frac{5}{6}} \beta \sqrt{S_n m^*} v_n^{rel} \quad (3)$$

The tangential contact force is expressed as follows:

$$F_t = \min\{\mu_s F_n, F_t^c + F_t^d\} = \min\left\{\mu_s F_n, -S_t \delta_t - 2 \sqrt{\frac{5}{6}} \beta \sqrt{S_t m^*} v_t^{rel}\right\} \quad (4)$$

where E^* , R^* and m^* are the equivalent Young's Modulus, equivalent radius, and the equivalent mass; δ_n and δ_t are the normal and tangential overlaps; v_n^{rel} and v_t^{rel} are the corresponding components of the relative velocity; μ_s is the coefficient of static friction; S_n and S_t are the normal and tangential stiffness, and are given by Eqs. (5) and (6);

$$S_n = 2E^* \sqrt{R^*} \delta_n \quad (5)$$

$$S_t = 8G^* \sqrt{R^*} \delta_n \quad (6)$$

where G^* is the equivalent Shear modulus. β is the damping coefficient, given by:

$$\beta = \frac{-\ln e}{\sqrt{\ln^2 e + \pi^2}} \quad (7)$$

where e is the coefficient of restitution. It is defined as the ratio of the velocity components along the normal plane of contact after and before the collision. And it is usually 0.6 according to experience and previous research (Zhang and Huang, 2022). The detailed values are listed in Table 2.

2.2. Model setup and calibration

To confirm that the input material property parameters and contact parameters can correctly simulate the flow-slit structure interaction, Jiang and Towhata (2013)'s small-scale flume test was employed as a benchmark, and the entire description is presented in Fig. 1. This test have been used as a benchmark by Huang et al. (2021), due to its detailed flow, impact and deposition characteristics data, showing its availability. In addition, there are many other tests that can be used as benchmarks due to the high-quality flow data, such as the granular chute experiments conducted by Pudasaini et al. (2007). A rigid barrier with a size of 0.40 m $\times$ 0.30 m was constructed at the flume's end. A total of 20,409 particles with diameters of 10 mm were generated,

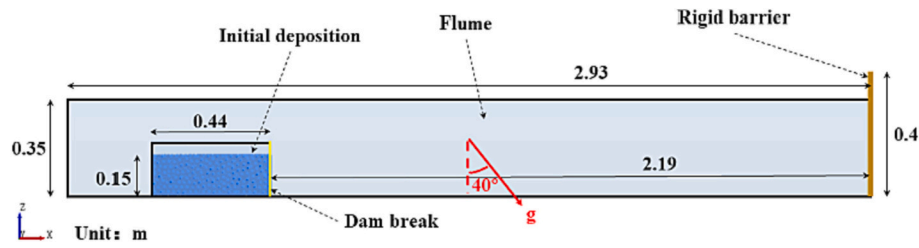


Fig. 1. Schematic diagram of DEM calibration from Jiang and Towhata (2013).

resulting in an initial deposition with dimensions of $0.15 \text{ m} \times 0.44 \text{ m} \times 0.30 \text{ m}$ and a unit weight of 13.5 kN/m^3 under gravity. Tables 1 and 2 outline the DEM's material property and contact parameters, which were chosen based on rock fragment characteristics. The choice of contact parameters also should consider the model size, the lower Elastic modulus might be more suitable for the small-scale DEM simulations (Goodwin and Choi, 2021). When all particles were set, the dam break was simulated by quickly turning the trigger gate. Then, the granular flow slid down the flume under gravity and impacted the barrier. Eventually, the granular flows were intercepted by the rigid barrier. The DEM model was calibrated by comparing the entire granular flow progress and normal impact load's time history curves, which were obtained by the flume test and DEM simulation, respectively.

Figure 2 shows the granular flow morphology at any time satisfactorily matches the flume test findings. And Fig. 3 depicts the temporal evolution of the impact force. As can be seen, the numerical data are consistent with the experimental data, particularly the peak impact force characteristics, except the residual forces. Although the residual forces are slightly larger compared with the flume tests results, the difference is acceptable because this parametric analysis did not focus on the impact force. As a result, the higher agreement between simulation and experiment in flow progress and impact force demonstrates that the DEM model can successfully describe the flow-structure interaction. However, It is worth noting that the impact force signal over time was greatly sensitive to the slope inclination of the flume, and it is necessary to investigate a much wider range of slope inclination to better calibrate a DEM model (Chanut et al., 2010).

2.3. Numerical simulation program

The numerical setup for the slit structures matched the geometry of the modeling flume used in the calibration stage, and only the closed barrier was replaced with an adjustable slit structure at the same position. Additionally, the height of the slit structure was the same as that of the closed barrier to avoid overtopping. The relative slit sizes (s/δ), defined as the ratio of slit width (s) and particle diameter (δ) varied from 2 to 10, which cover the entire range of the existing guidelines, e.g. China (MLR, 2004) and Japan (NILIM, 2005). The relative slit separation (L/δ), defined as the ratio of slit separation (L) and particle diameter (δ) varied from 1 to 10, the number of slits (N_{sl}) was one or two, the slope inclination (θ) varied from 30° to 45° . Table 3 summarizes the simulation

Table 1
Properties of materials.

Material	Density $\rho_s(\text{kg/m}^3)$	Poisson's ratio ν	Young's modulus $E(\text{MPa})$	Source
Particles	2550	0.25	100	Limestone (Shen et al., 2018)
Flume	7900	0.3	2×10^3	Steel material (Huang et al., 2021)
Barrier	7900	0.3	2×10^3	Steel material (Huang et al., 2021)

Table 2
Contact parameters.

Parameter	Particle- Particle	Particle- Flume base	Particle- Slide wall	Particle- Barrier	Source
Contact model	Hertz-Mindlin (no slip)				Tsuji et al. (1992)
Coefficient of restitution e	0.6	0.6	0.6	0.6	Jiang et al. (2018)
Friction coefficient μ_f	1.33	0.466	0.268	0.384	Measured by Jiang and Towhata (2013)
Rolling friction coefficient μ_r	0.06	0.01	0.01	0.01	

plan of DEM. The slit width (s), slit separation (L) and slit numbers (N_{sl}) are all shown in Fig. 4. The monitoring sections I are put before the slit structure to capture the flow kinematics. The monitoring sections span both the whole width (0.3 m) of the flume and the entire height (0.40 m) of the sidewalls to guarantee that no airborne particles are omitted. Goodwin (2019) conducted a sensitivity study and found that 5 particle diameters is a proper thickness value. Thus, the thickness of monitoring sections is also set to 50 mm, which is 5.0 times the particle sizes used in the simulation (10 mm).

3. Results

3.1. Dynamics of flow-slit structure interaction

First, the overall interaction and impact process of the granular flows through the slit structures were observed and then compared with those of the closed check dam, as shown in Fig. 2. Fig. 5 shows a side view of the observed flow kinematics of dry granular flows against (a) single-slit structure (test N1-S3, wherein $N_{sl} = 1$, $s/\delta = 3$); (b) dual-slit structure (test N2-S3-L4, wherein $N_{sl} = 2$, $s/\delta = 3$, $L/\delta = 4$). For the closed channel condition (Fig. 2), the granular flows reached the rigid barrier with a very thin and dilute front (Fig. 2 (b3)). Then, the granular flow's front part impacted the rigid barrier and partly rebounded, and majority of the granular materials was intercepted, forming a dead zone (Fig. 2 (b4)). As the subsequent flow arrived at the rigid barrier, it was buried, moved on the settled sediments, and impacted the upper rigid barrier (Fig. 2(b5)) until the entire impact finished (Fig. 2 (b6)), which is consistent with Albaba et al. (2015)'s experimental and simulations results.

For the single-slit structure (Fig. 5 (a)), the granular flows reached the slit structures with the same state because they had the same initial modeling configuration (Fig. 5 (a1)). Then, the front part of the granular flow impacted the slit structures and most granular material was held in front of the rigid barriers on both sides, while a part of the granular material passed through the aperture (Fig. 5 (a2)). The pile-up process was constantly developed as the successive flow glided over the settled

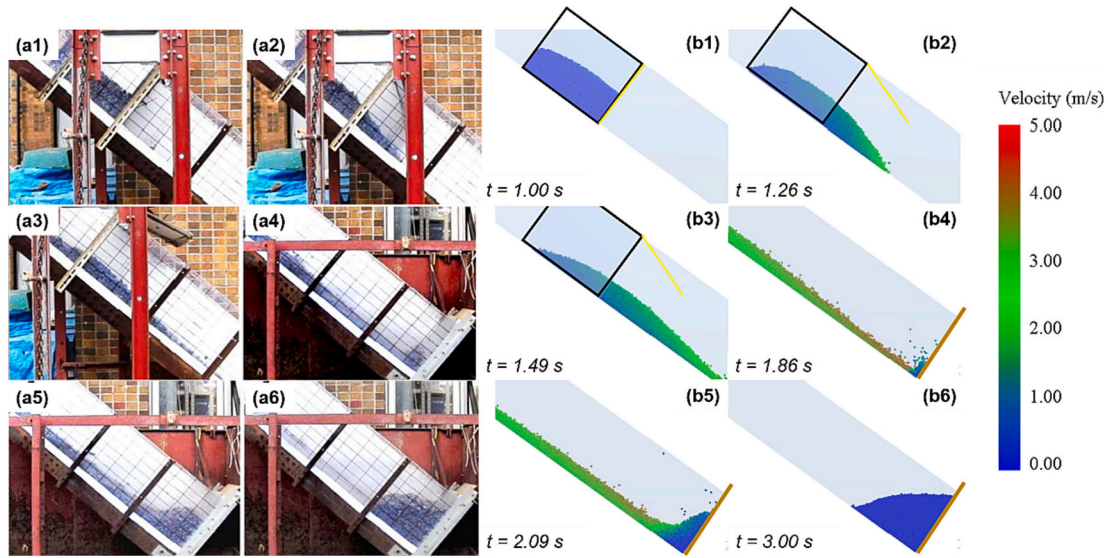


Fig. 2. Impact process of granular flow obtained from (a) flume tests in Jiang and Towhata (2013) and (b) DEM simulation in this study.

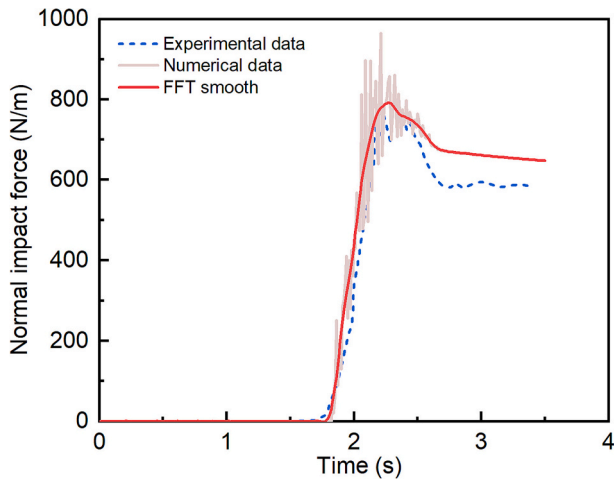


Fig. 3. Evolutionary behaviors of impact force obtained by flume tests in Jiang and Towhata (2013) and DEM simulation in this study.

Table 3

Numerical model plan of flow-slit structure interaction.

Test series	Slope angle	Numbers of slit N_{sl}	Relative Slit separation (L/δ)	Relative slit size (s/δ)
Open channel	30°, 35°, 40°, 45°	1	—	30
Closed channel	30°, 35°, 40°, 45°	—	—	—
Single-slit	30°, 35°, 40°, 45°	1	—	2, 3, 4, 5, 6, 8, 10
Dual-slit	30°, 35°, 40°, 45°	2	1, 2, 3, 4, 6, 8, 10	2, 3, 4, 5, 6, 8, 10

The “—” symbol indicates that the influence of this factor is not considered.

sediments (Fig. 5 (a3)), until the dead zone reached the maximum height. Subsequently, a part of trapped granular matter slowly flowed through the mouth due to the break of clogging granular arches under the gravity (Fig. 5 (a4)), which reduced the height of the dead zone and finally formed a ramp-like dead zone whose height was reduced from the flume sidewall to the mouth (Fig. 5 (a5) and Figs. 6–7). When the

granular flows impacted the dual-slit structure (Fig. 5 (b)), the entire impact process was approximately the same as that of the single-slit structure in the side view. However, there are still some differences, which were caused by the different granular arches, as will be explained below.

3.2. Granular arches in multi-slit structures

Although the richness of the evolution behaviors of the granular arches can be observed, previous studies rarely considered this clogging evolution mechanism and instead focused on the macroscale impact. The underlying mechanisms controlling these surface phenomena are discussed in this section.

Figures 6 and 7 show the final deposition spatial morphology of dry granular flow behind the single-slit and dual-slit structure in the form of particle and force chains from front and top view, respectively. For the single-slit configuration with $s/\delta = 2$ (Fig. 6 (a1)), the stable granular arch formed, owing to the narrower aperture. As s/δ increases, more granular matter is required to form the granular arch until $s/\delta > 6$ (Fig. 6 (b1–d1)). Meanwhile, there is a certain degree of increase of the frontal velocity because of the streamlines contraction in penetrating zone between the dead zone (Rossi and Armanini, 2019; Zhou et al., 2020), as shown in Figs. 6(a1–d1) and 7(a1–d1), leading to a stronger impact force on the granular arches. Thus, the formed arch may easily be destroyed by the subsequent granular flows.

As for the dual-slit structure, owing to the generating zone overlaps of the two nearby slits, the shearing effect of the particles between the two slits affects the stability of each slit's clogging granular arch structures, which has been discussed by Goodwin and Choi (2020). More granular flows will simultaneously pass through the slits until stable granular arches are formed. However, with the increase of L/δ , the generating zones of the two near slits gradually separate such that the dual-slit structures are simply the accumulation of each individual slit, as shown in Fig. 7(c2–d2).

4. Regulation function of dual-slit structures

In this part, the pile-up height, mass outflow rate, grain-trapping efficiency, and energy breaking capacity are analyzed to systematically investigate the functions of multi-slit structures for intercepting and regulating rapid dry granular flows, comparing with single-slit structures.

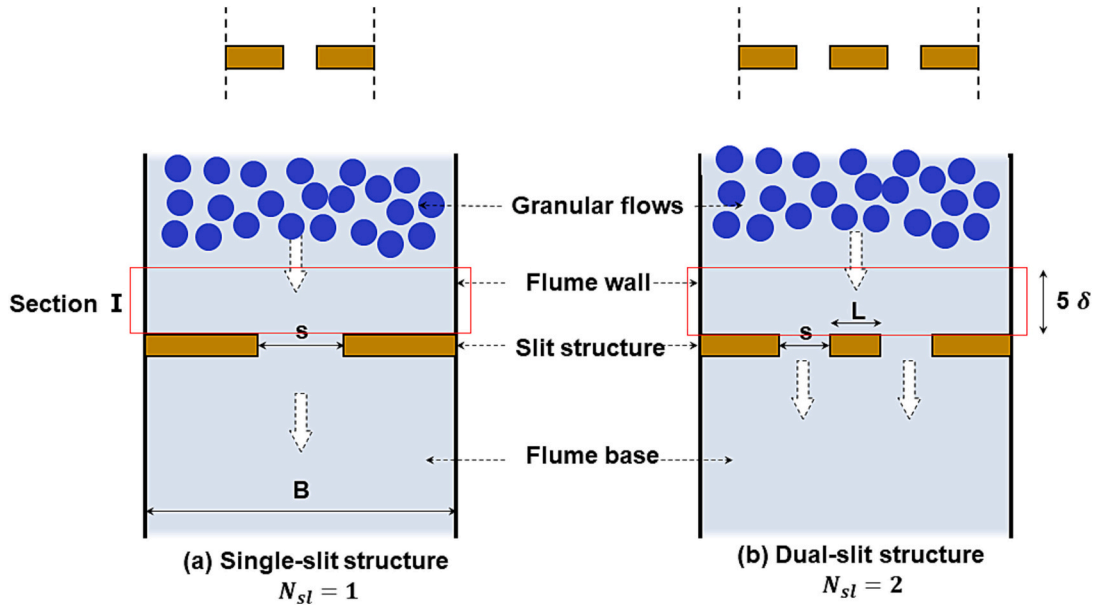


Fig. 4. Top view of (a) single-slit structure and (b) dual-slit structure. Granular flow passes through the slit structure from above. In this snapshot, s is the slit width, L is the slit separation, N_{sl} is the number of slit, and B is the width of flume channel. The monitoring section I, represented as a red rectangle span both the whole width (0.3 m) of the flume and the entire height (0.40 m) of the sidewalls to guarantee that no airborne particles are omitted. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.1. Influence of multiple slits on pile-up height

To prevent overtopping, the pile-up height must be estimated when designing barrier countermeasures (Ng et al., 2017; Zhou et al., 2020). In this DEM model, section I was set before the slit structures (Fig. 4) to measure the pile-up height. The pile-up height was calculated by Eq. (7):

$$h = \frac{2}{N} \sum_{i=1}^N h_i \quad (7)$$

where N is number of particles in section I at any moment; h_i is the height of the center of the i^{th} grain.

Figure 8 shows the historic characteristic curves of the pile-up height normalized by heights of incoming flows $h_i = 30.7\text{mm}$ for three scenarios: closed channel, single-slit structure, and dual-slit structure with different slit sizes. The time is normalized by free fall time $\sqrt{H/g}$, and the initial time is set 0 through subtracting $t_0 = 1.75\text{ s}$. In each case, the pile-up height rapidly reaches its peak values with the same trend, then slowly decreases, and finally maintains a constant value. For the closed channel, the normalized pile-up height h_f/h_i reaches a peak of 10.13 at $[t] = (t - t_0)/\sqrt{H/g} = 2.97$. For $s/\delta = 2$ with either a single slit or dual slits, the normalized pile-up height also reaches a peak of approximately 10.11 at $[t] = 2.97$, and then there is no more granular flows through the outlets. This suggests that the slit structure with $s/\delta \leq 2$ will not play a regulation function, like a closed check dam. This conclusion is consistent with Zhou et al. (2019)'s findings.

Figure 9 shows the effects of the relative slit size s/δ on maximum pileup height h_f normalized by heights of incoming flows h_i . For both single and multi-slit structures, the numerical results show a non-linear connection between h_f/h_i and s/δ . The change of the normalized maximum pileup height h_f/h_i is not significant when $s/\delta < 4$, and it rapidly decreases for $s/\delta > 4$, for single-slit structure and multi-slit ones with wider slit separation. The critical value has been widely discussed before. The reason for this is that the stable granular arches are easier to develop when $s/\delta < 4$ for both single-slit structure and multi-slit ones with wider slit separation. It is worth noting that, as shown in Fig. 9, the maximum pileup height of multi-slit structures with narrow slit separation is lower than multi-slit structures with wider slit separation. The

reasons for this may be that the shearing effect of the particles between the two nearby slits affects the stability of each slit's granular arch structures, making difficulty in forming the stable granular arches. This weakening effect of narrow slit separation overcomes the clogging effect of stable granular arches, thus, the maximum value is reduced. This implies that the slit separation L/δ , which often has been ignored in previous designs, also play a vital role in slit-structure performance.

4.2. Influence of multiple slits on mass outflow rate

The most important functions of slit structures are to reduce the peak flow rate. Therefore, this study performed parametric analysis on the mass flow rate. The evolution of the mass outflow rate with varied slit sizes and slit separation is shown in Fig. 10(a). In each case, normalized mass flow rate rapidly reaches its peak values at the same time; then, the rate decreases and eventually decreases to zero for a long time. With wider relative slit sizes, the maximum normalized mass flow rate becomes larger. For a given s/δ , with regard to the single-slit and dual-slit structure, the maximum value of the latter is approximately twice that of the former. The reason for this may be that the relative slit separation L/δ is sufficiently far and the adjacent granular arches do not interfere with each other, thus generating two stable granular arches, leading to the normalized mass flow rate of dual-slit structure is approximately twice that of the single-slit structure.

The Beverloo law has been widely used in hopper flow (Alonso-Marroquin and Mora, 2020; Thomas and Durian, 2013; Zhang et al., 2016), and further be modified by Goodwin and Choi (2020) for channelized flows through some adaptive improvement, which has provided an excellent estimate. However, there is a certain degree of increase of the frontal velocity because of the streamlines contraction by granular dead zone (Rossi and Armanini, 2019; Zhou et al., 2020), which has not been considered in outflow rate before. Here, we introduce the contraction coefficient $\alpha = \frac{u_z}{u_0} = \left(\frac{s}{B}\right)^e$ proposed by Zhou et al. (2020) to consider the contraction behaviors near slits. The further modified Beverloo law with the α is then given by Eq. (8):

$$\dot{M} = k_1 N_{sl} v_s \rho_s (s - k_2 \delta) \cdot \alpha h U_0 \quad (8)$$

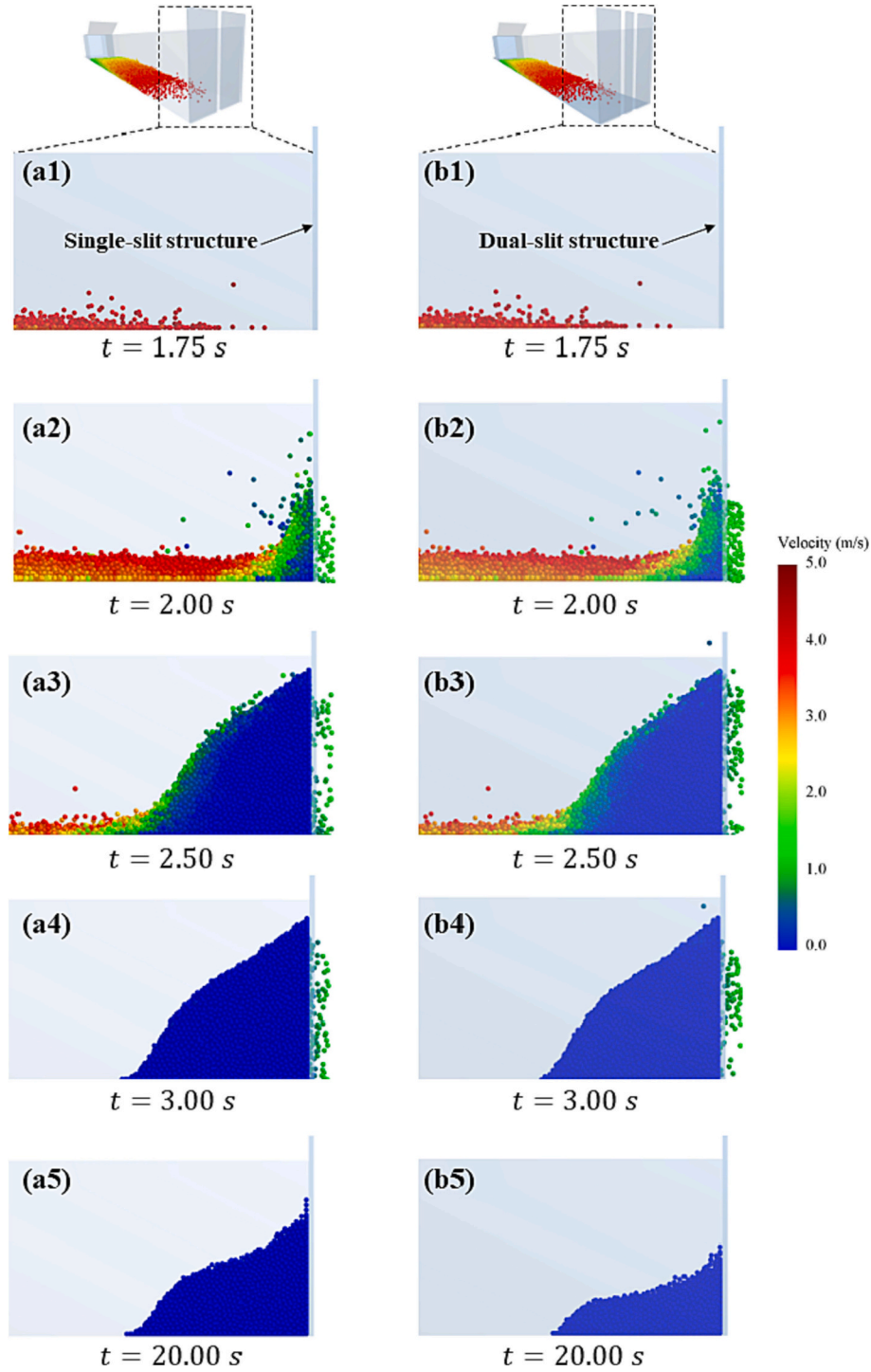


Fig. 5. Right side view at the end of slit structure of the observed flow kinematics of dry granular flows against (a) single-slit structure (test N1-S3, wherein $N_{sl} = 1$, $s/\delta = 3$); (b) dual-slit structure (test N2-S3-L4, wherein $N_{sl} = 2$, $s/\delta = 3$, $L/\delta = 4$).

where $\dot{M}$ is the mass flux; k_1 and k_2 are parameters; v_s is the particle volume fraction; ρ_s is the particle density, h is the flow depth, U_0 is the flow velocity.

Thus, the mass outflow rate normalized by peak mass flow rate of the open channel is:

$$\dot{M}_{norm} = \frac{\dot{M}}{\dot{M}_{open}} = \alpha \frac{N_{sl}(s - k_2\delta)}{(B - k_2\delta)} \quad (9)$$

where the B is the channel width.

The normalized peak outflow rate predicted by Goodwin and Choi (2020) and the further modified formula considering contraction behaviors near slits in this work are shown in Fig. 10(b). It is noted that, although the results predicted by Goodwin and Choi. (2020) and this study show a good prediction ability, the method in this study considering the contraction behaviors showing an exponential relationship provides a better prediction. We also can conclude that the relative slit separations have a negligible effect on mass outflow rate.

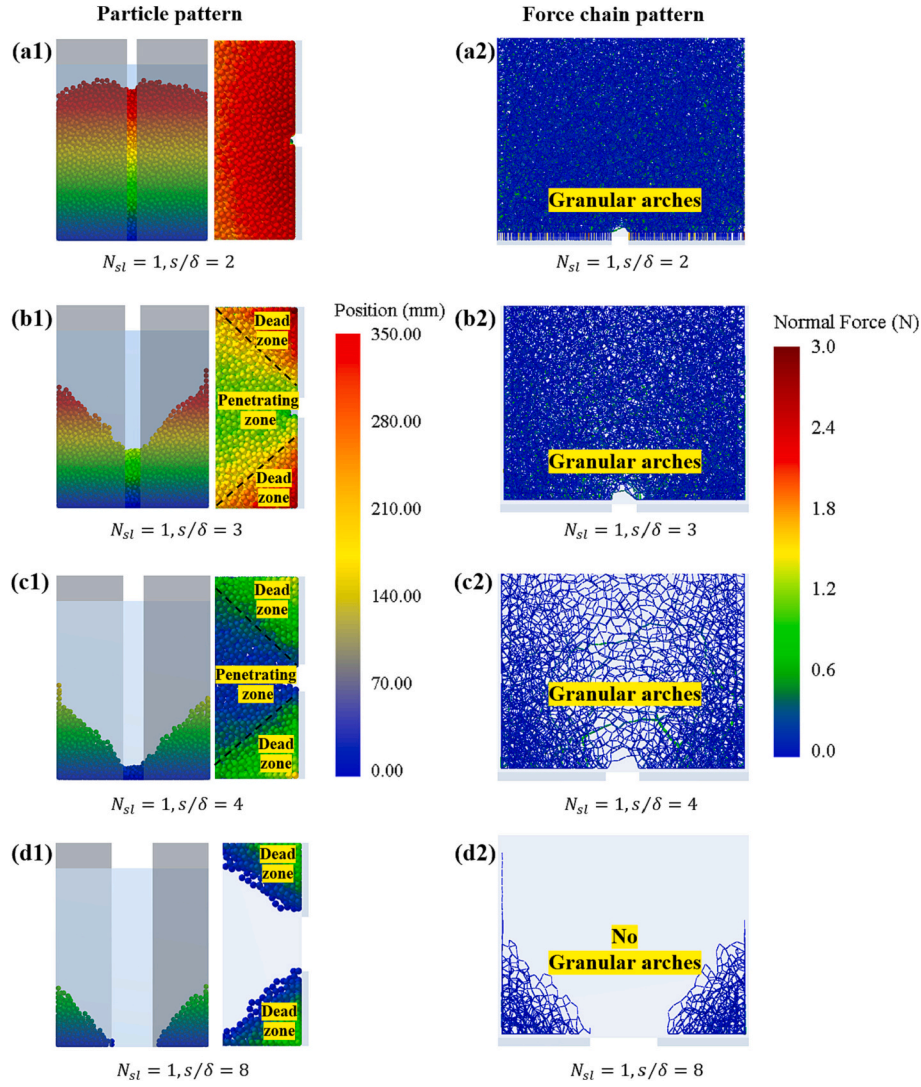


Fig. 6. Final deposition spatial morphology of dry granular flow behind a single-slit structure. (a1-d1) Particle clogging arches from front and top view; (a2-d2) Force chains from top view.

4.3. Influence of multiple slits on grain-trapping efficiency

It is crucial to achieve a balanced trapping efficiency in designing slit structures (Gong et al., 2021). This study defines the grain-trapping efficiency as follows:

$$E_G = \frac{N_{trapped}}{N_{total}} \quad (10)$$

where $N_{trapped}$ is the number of the particles trapped by the slit structures, and N_{total} is the total amount of simulated particles in this event. As E_G becomes larger, the trapping capacity of the slit structure for that configuration improves, with unity indicating complete trapping.

The evolution of grain-trapping efficiency with various slit sizes is shown in Fig. 11. In each case, as s/δ increases the grain-trapping efficiency first remains approximately constant ($>90\%$) when $s/\delta \leq 2$, and then rapidly decreases to $<20\%$ when $s/\delta \geq 4$. These thresholds are the same for both single-slit structures and multi-slit structures. These trends are highly consistent with the normalized maximum pileup height. The reason for this may be the fact that, when the aperture is very small ($s/\delta \leq 2$), clogging occurs quickly and only a few particles pass through the outlets. In contrast, when the aperture is very wide ($s/\delta \geq 4$), a stable clogging granular arch hardly forms and the slit structures are no longer working; resulting the lower grain-trapping efficiency.

Additionally, when $2 < s/\delta < 4$, as the width of the slit increases, the formation of the granular arch becomes more difficult. This implies that the formation and break of the granular arch controls the grain-trapping efficiency.

The evolution of grain-trapping efficiency with various slit separations for different slope inclination is shown in Fig. 12. As shown in the Fig. 12(a-c), it was found that, there is an obvious peak value when $L/\delta = 4$. When $L/\delta \leq 4$, the grain-trapping efficiency increases with increasing slit separations, and gradually falls for $L/\delta > 4$. As mentioned in the “pile-up height” section, the reasons for this may be that both shearing effect of the particles between the two nearby slits and the forming of the stable granular arches together control the grain-trapping efficiency. Furthermore, we can conclude that the Froude characteristics also influence this peak value. For a larger given Froude number (see Fig. 12(d)), there are no peak value within our study range. Thus, it remains to be further investigated about rational design of grain-trapping efficiency by considering slit sizes, slit separations and slope inclinations.

4.4. Influence of multiple slits on energy-breaking efficiency

The kinetic energy of the particles within monitoring section I are calculated by Eq. (11):

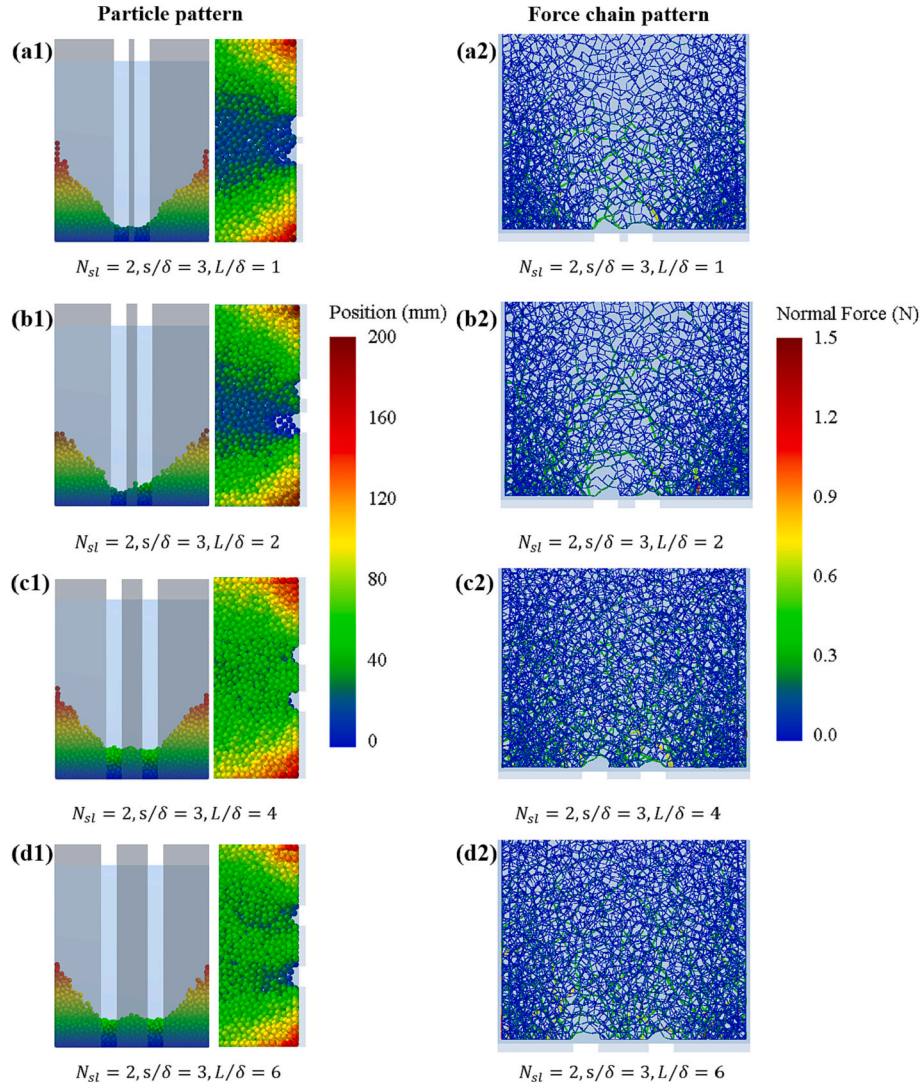


Fig. 7. Final deposition spatial morphology of dry granular flow behind a dual-slit structure. (a1-d1) Particle clogging arches from front and top view; (a2-d2) Force chains from top view.

$$K = \sum_{i=1}^N \left(\frac{1}{2} m_i v_i^2 + \frac{1}{2} I_i \omega_i^2 \right) \quad (11)$$

where N is the number of particles within monitoring section I at a given time.

Figure 13 shows the granular flows' kinetic energy for different (a) s/δ and (b) L/δ . In each case, it was found that the kinetic energy rapidly reaches its peak values, then decreases, and eventually becomes zero for a relatively long time. The time for reaching the peak values increases with the relative slit size s/δ . Additionally, as s/δ becomes wider, the kinetic energy becomes larger. Moreover, it was concluded that the slit separation has no effect on the kinetic energy. To accurately quantify the energy dissipation ability of slit structures, the energy-breaking efficiency is defined as follows (Marchelli et al., 2020):

$$E_k = \frac{\int_{t_0}^t (K_{open} - K_{slits}) dt}{\int_{t_0}^t K_{open} dt} \approx \frac{\sum_{t_0}^t K_{open} - \sum_{t_0}^t K_{slits}}{\sum_{t_0}^t K_{open}} \quad (12)$$

where t_0 is the moment when the first impact occurs; here, $t_0 = 1.75$ s and $t = 3.00$ s. K_{open} is the kinetic energy when no barrier is present, and K_{slits} is the kinetic energy when slit structure is present. The energy-

breaking capacity as function of s/δ and L/δ is presented in Fig. 14, all of results of this study including single-slit and multi-slit structures are represented as solid lines, while the single-slit results for different Froude numbers from Marchelli et al. (2020) are represented as dotted lines. It should be pointed out that, the Froude characteristics for channel inclination 40° , here $Fr = \frac{v_f}{\sqrt{g h \gamma \cos \theta}} = 5.86$ is slightly larger than Marchelli et al. (2020). It is apparent that both single-slit and multi-slit structures has excellent energy-breaking ability, with values $>70\%$. For a given value of s/δ , the energy-breaking efficiency K might be controlled by Fr for the lower Fr , while be irrelevant when Fr is larger. Moreover, for both single-slit and multi-slit structures, Fig. 14(a) reveals a near-linear connection between energy-breaking efficiency and slit size. The only difference lies in the sensitivity to s/δ . With the increase of s/δ , energy-breaking efficiency of multi-slit structures fall faster than the single-slit ones. This is well understood, the transverse blockage $(B - s)/B$ of multiple ones should multiply slit numbers N_{sl} , showing a faster increase than the single ones. Moreover, Fig. 14(b) shows that the slit separation has a negligible effect on energy-breaking efficiency.

5. Discussion

It is to be noted that, debris flows are poorly sorted and water-rich

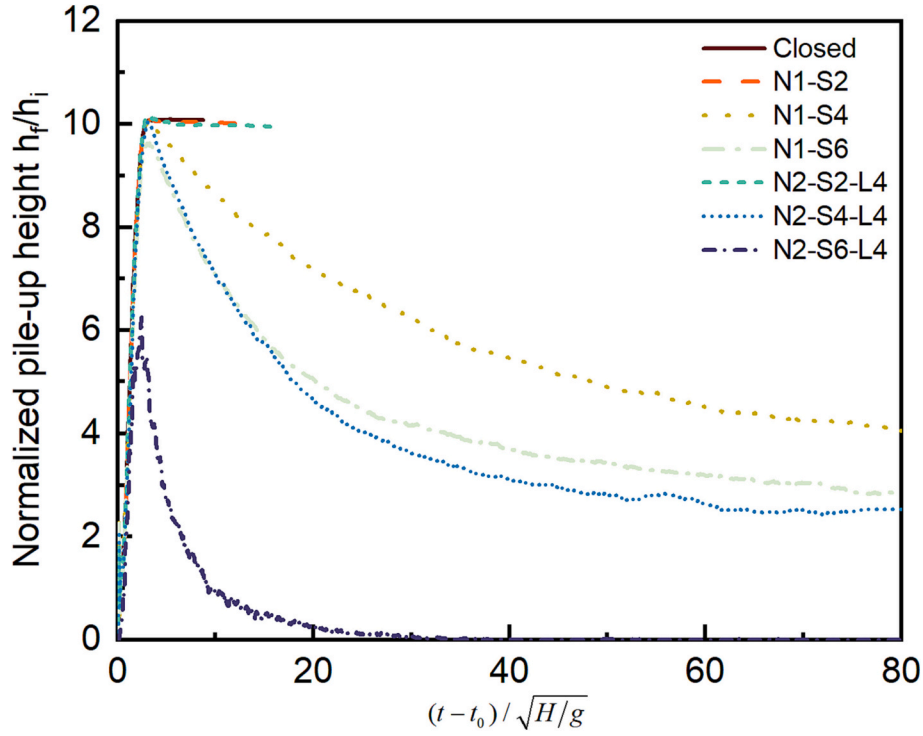


Fig. 8. Historic characteristic curves of pile-up height for three scenarios: closed channel, single-slit structure, and dual-slit structure with different slit sizes.

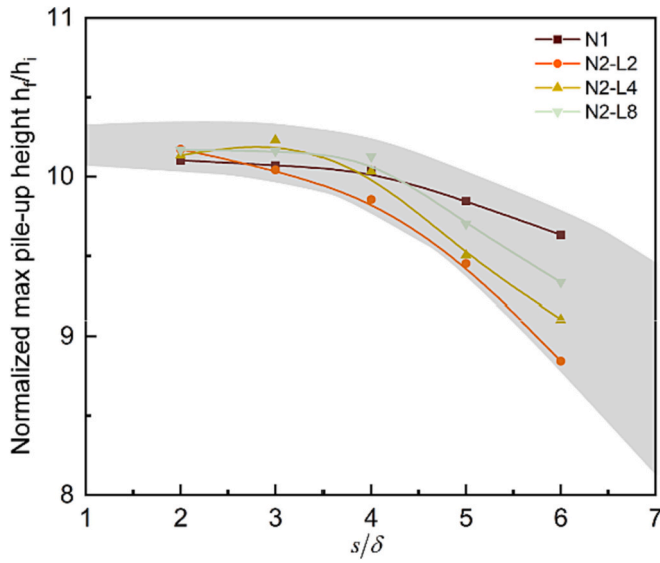


Fig. 9. Effects of relative slit size s/δ on normalized maximum pile-up height.

(Gray and Ancey, 2011; Iverson et al., 2016; Jiang and Zhao, 2015; Ng et al., 2017; Song et al., 2017). Engineering design of multi-slit structures, such as regulation functions that take into account particle shape and the absence of a pore fluid, is difficult and requires more research. In this study, dry granular flows were modeled as an ensemble of rigid monodisperse spheres. Although we have included a rolling friction torque to compensate the simplification of particle shape, purely spherical particles indeed cannot represent the real irregular granular flows (Zhang et al., 2020). This might be a poor choice, especially for the flow-slit interaction, resulting in weaker granular arches and more outflow rate (Cagnoli, 2021; Marchelli et al., 2020; Zhou and Sun, 2013). Furthermore, the non-spherical particles also has been used in channel flows (Albaba et al., 2015). To progress on this issue, the non-

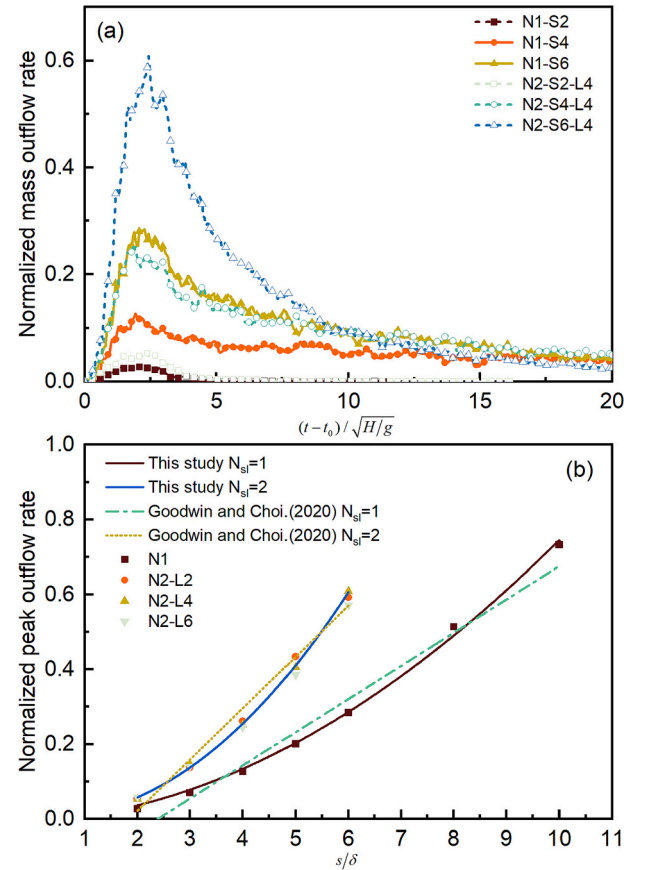


Fig. 10. (a) Comparison of normalized mass flow rate in single-slit structure and dual-slit structure, (b) The normalized peak outflow rate predicted by Goodwin and Choi. (2020) and this study considering contraction behaviors near slits.

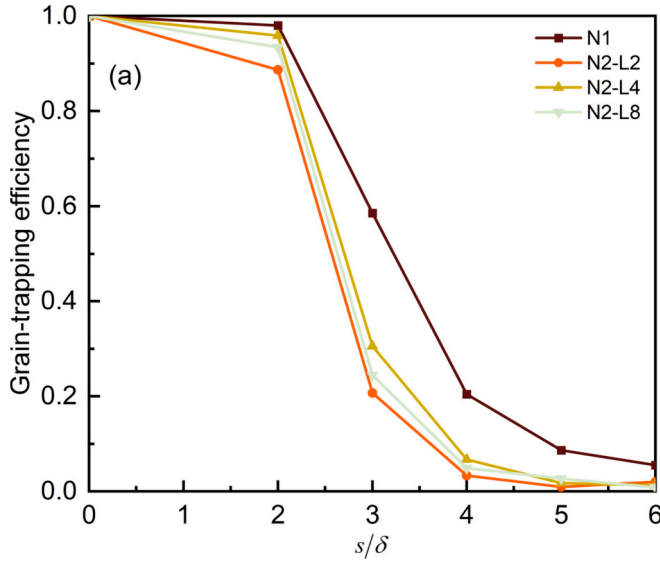


Fig. 11. Grain-trapping efficiency as function of s/δ .

spherical particles should be used in flow-slit interactions in the future.

In terms of the fluid effect, the particle–fluid interaction in debris flow may reduce particle contact friction through buoyancy forces and lubricating effect, therefore, significantly enhance the mobility (Fang et al., 2021). Many efforts have also been devoted to focus on this issue, for example, Choi et al. (2015a, 2015b) dictates the fluid phase

significantly influences the impact mechanism. Song et al. (2017) firstly demonstrated that a higher solid fraction results in a higher impact force. However, to our knowledge, few simulation studies have been done to investigate the influence of particle–fluid interaction on regulation function of slit structures. Therefore, particle morphology and fluid effect should also be investigated in future work.

6. Conclusions

The main body of this study was to systematically investigate the functions of multi-slit structures for intercepting and regulating rapid dry granular flows, comparing with single-slit structures. In this study, DEM was adopted to investigate the flow-slit structure interaction mechanism and the regulation functions for the pile-up height, mass outflow rate, grain-trapping efficiency, and energy-breaking capacity by altering the slit sizes, slit separation and slit numbers. The following conclusions can be drawn as follows:

1. Combining numerical results and previous observations reported in the literature, the dynamics process of flow-slit structure interaction and the effects of granular arches in multi-slit structures has been explicitly analyzed. It's worth noting that, the streamlines contraction behavior by granular dead zone plays a key role in slit structures' regulation function for mass outflow rate. The contraction coefficient is introduced to further modify Beverloo law. This method provides a good prediction to the exponential relationship.
2. The simulation results imply that, the s/δ and N_{sl} significantly influence the pile-up height, mass flow rate, grain-trapping efficiency, and energy-breaking capacity. As s/δ becomes wider, h_{max} , E_G , and

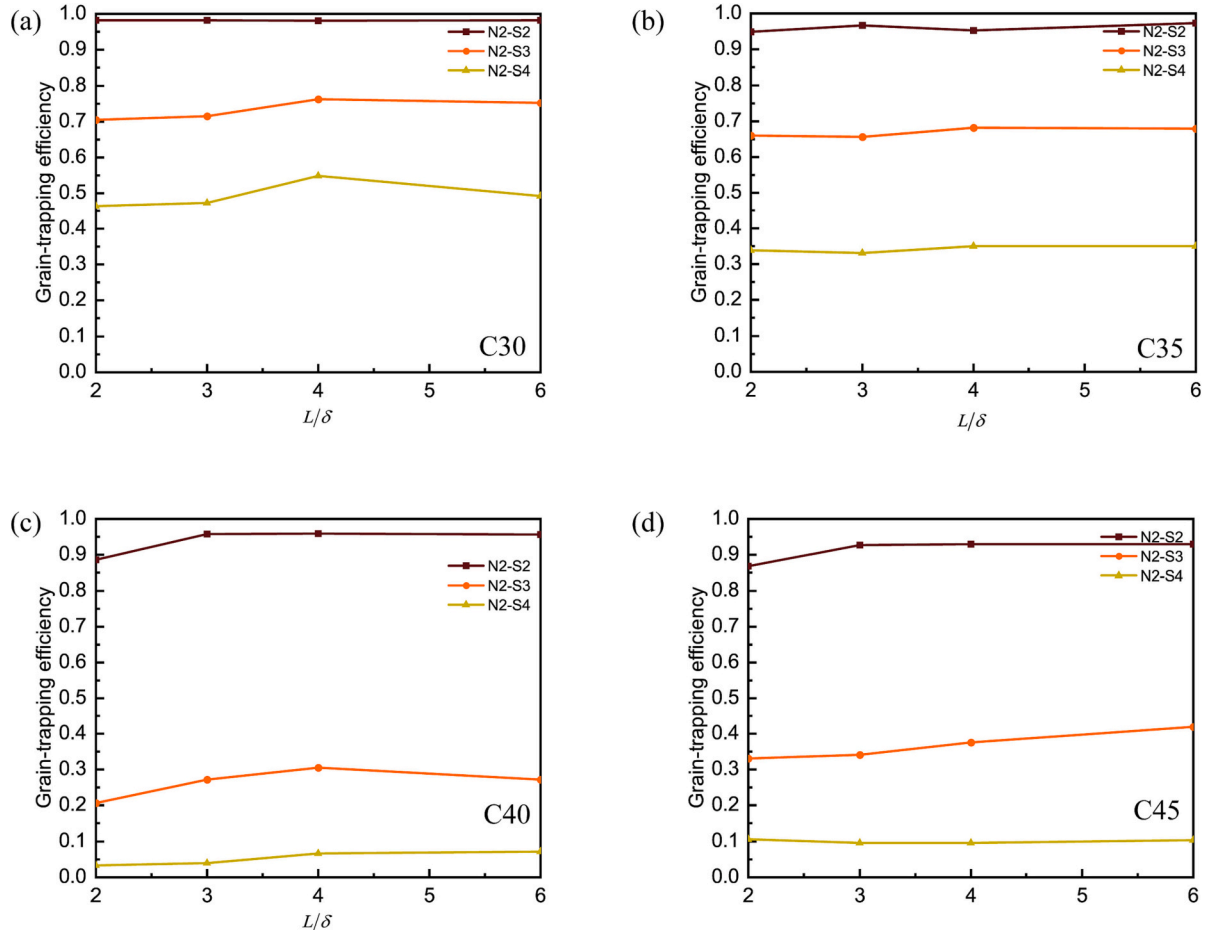


Fig. 12. Grain-trapping efficiency as function of L/δ for different slope inclination: (a) 30°, (b) 35°, (c) 40°, (d) 45°.

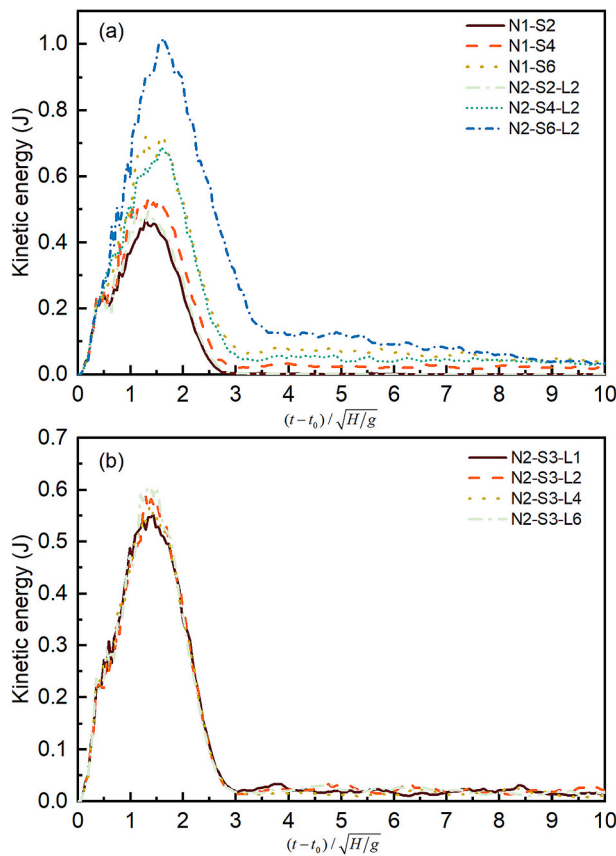


Fig. 13. Kinetic energy of granular flows for different (a) s/δ and (b) L/δ .

even E_k decrease. However, the maximum normalized mass flow rate is positively correlated with s/δ , as is also the case for N_{sl} .

3. Compared to single-slit structures, the slit separation L/δ also serves a non-negligible role in multi-slit-structure. For dry granular flow with a lower Froude number (5.86, in this study), when the relative slit separation $L/\delta \leq 4$, the shearing effect of the particles between the two nearby slits might overcomes the clogging effect of granular arches, making a lower grain-trapping efficiency. When the relative slit separation $L/\delta > 4$, no significant influence was observed. For dry granular flow with a larger Froude number, there are no peak value within our study range. Thus, it remains to be further investigated about rational design of grain-trapping efficiency by considering slit sizes, slit separations and slope inclinations.

All in all, although the multi-slit structure is a more effective countermeasure for regulating the debris flows compared with single-slit structures, rational designs of structural parameters, such as slit sizes and slit separation should be pay more attention. Despite the limitations of simulation model configurations, the findings of this study can serve as a basis for the scientific design of multi-slit structures.

Data availability statement

All data, models, and code generated or used during the study appear in the submitted article.

CRediT authorship contribution statement

Yu Huang: Conceptualization, Methodology, Project administration.
Yandong Bi: Visualization, Writing – original draft, Writing – review & editing.
Bei Zhang: Investigation, Formal analysis.

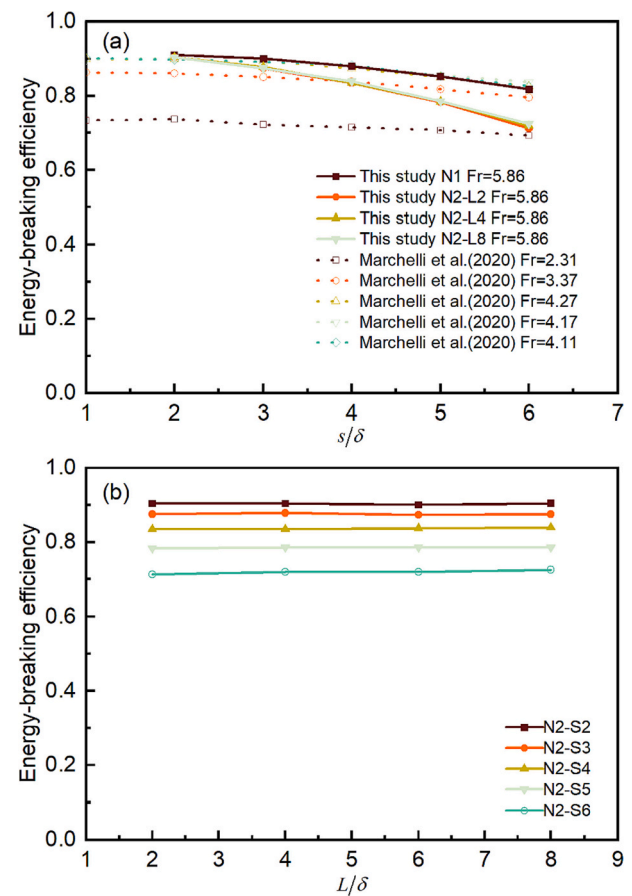


Fig. 14. Energy-breaking capacity as function of (a) s/δ and (b) L/δ .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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